

Casey D. Foley, PhD

Laser & Vacuum Processes | Optical Metrology & Machine Vision | Spectroscopy | Process Development

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SUMMARY

PhD physical chemist and Staff Scientist with 12+ years across production manufacturing, a national laboratory, and university research. Develops and scales laser and high-vacuum processes, with results of 99% yield, a 3× takt-time reduction, and 40% faster evacuation. Builds production metrology from algorithm to plant floor, including laser-based R-value measurement, phase-shifting deflectometry, and confocal profilometry, and manages a team of three manufacturing engineers. 23 peer-reviewed publications, including *Science* and *JACS*.

CORE SKILLS

Lasers & optics	Fiber, Nd:YAG/DPSS, CO ₂ , excimer, Ti:sapphire, dye, IR OPO/OPA, VUV; ablation, scribing, laser heating; optical alignment, beam diagnostics, Zemax OpticStudio, laser safety
Vacuum & instruments	UHV design and construction, production high-vacuum processes, cryogenics, ion optics, high voltage, molecular beams, particle imaging, SolidWorks/FreeCAD
Metrology & vision	Laser-based R-value measurement, phase-shifting deflectometry, confocal profilometry, thermal and optical-distortion scanning, machine-vision defect detection, color measurement, ML surface inspection
Spectroscopy	UV/Vis/IR, FTIR, REMPI, LIF, IR-UV double resonance, velocity-map imaging, mass spectrometry (Waters, Thermo, Bruker), ion mobility
Process & quality	Process development and scale-up, DOE, SPC, Cp/Cpk and Pp/Ppk, PFMEA, 8D root-cause analysis, PDCA, control plans, SOPs, preventive maintenance, KPIs, Lean and Six Sigma training, team leadership
Software & data	Python, LabVIEW, MATLAB, data acquisition and instrument control, machine learning, statistical analysis, automation

PROFESSIONAL EXPERIENCE

Staff Scientist	Apr 2026 – Present
Laser Systems Engineer	Aug 2024 – Mar 2026
<i>LuxWall, Inc. — vacuum-insulated glass manufacturing</i>	
Integrated a fiber-laser ablation system into production, cutting scrap from 1 in 9 parts to 1% (99% yield) with a 3× takt-time reduction. Brought a second ablation process to 99% yield with a 2× takt reduction through sensor integration and statistical process control. Led development of the vacuum-insulated-glass (VIG) evacuation process for the company's first factory at 99% yield, and reduced evacuation takt time 40% while maintaining unit quality. Developed a proprietary laser-based R-value measurement system from the measurement algorithm through plant-floor integration, and the protocol for using it in production quality screening. Led flatness and distortion metrology: developed an offline confocal gantry profiler with Python control and analysis (micron level) and an in-line phase-shifting deflectometry camera system (~10 μm), and wrote analysis software to extract process parameters from distortion tools and a thermal scanner. Manage a team of three manufacturing engineers (since Apr 2026). Subject-matter expert for all laser and high-vacuum processes and technical liaison between factory engineering, scaling, and R&D; implemented laser-safety solutions, authored SOPs, work instructions, control plans, and troubleshooting guides, and overhauled preventive maintenance. Commissioned equipment and supported the launch of an automated production line, working with controls engineers on integration. Wrote requirements and specifications, led design reviews, and selected vendors for the machine-vision, R-value, and flatness measurement systems.	

- **Deployed machine vision and sensors for quality control:** brought a single-pane glass defect-detection system (50 μm resolution) into production use with a vision-system integrator, and added material color measurement and ML-based surface inspection.
- **Built a root-cause-analysis platform** that improved yield and made production data available company-wide.
- **Led development of laser-heating processes** for VIG production.

Research Excellence Postdoctoral Fellow

Aug 2022 – Jul 2024

University of Missouri — Department of Chemistry

- **Designed and built a custom UHV imaging instrument** with electron-ion coincidence detection for hypersonic collision studies.
- **Implemented a laser system for VUV detection** of atoms and molecules, generating 118 nm light by frequency-tripling 355 nm in a xenon gas cell. Performed MeV ultrafast electron diffraction experiments at SLAC National Accelerator Laboratory, with custom Python analysis.
- **Senior postdoc:** mentored and trained group members, and led lab operations and instrument training.

Postdoctoral Associate

Mar 2021 – Jul 2022

Sandia National Laboratories — Combustion Research Facility

- **Restored a \$1M+ cryogenically cooled ion-trap mass spectrometer:** rebuilt the UHV system, installed a tunable IR OPO/OPA laser, and programmed the LabVIEW instrument-control and data-analysis software.
- **Developed IR, UV, and IR-UV double-resonance spectroscopy methods** for molecular detection and structural characterization of metal-molecule complexes. First-author results appeared in *JACS* (2024) and *J. Phys. Chem. A* (2023).

Graduate Research Assistant (PhD)

May 2016 – Dec 2020

University of Missouri — Suits Group

- **First-author Science paper (2021)** on quantum resonances in formaldehyde photodissociation, measured by velocity-map imaging.
- **Designed laser automation systems and custom optical detection** for UHV molecular-beam experiments.

Graduate Research Assistant

Jun 2014 – Apr 2016

Wayne State University

- **Developed mass spectrometry methods** on Waters, Thermo, and Bruker platforms, including ion mobility, for biological and polymer analysis.

EDUCATION

PhD, Physical Chemistry, University of Missouri — *Quantum Aspects of Roaming Dynamics* (Prof. Arthur G. Suits) Dec 2020

BS, Biochemistry, Saginaw Valley State University — President's Scholarship; toxicology co-op at The Dow Chemical Company (2010–2012) May 2013

Honors: Army Research Laboratory Journeyman Fellowship (2018) · WSU Thomas C. Rumble Fellowship (2014–2015) · WSU Chemistry Citation for Excellence in Teaching (2014) · SVSU President's Scholarship (2009–2013)

SELECTED PUBLICATIONS

23 peer-reviewed papers (8 first-author) and a 2026 preprint · 450+ citations · h-index 10 · full list at cdfoley.com/publications

- **C. D. Foley**, C. Xie, H. Guo, A. G. Suits, "Orbiting resonances in formaldehyde reveal coupling of roaming, radical, and molecular channels," *Science* 374, 1122–1127 (2021).
- **C. D. Foley** et al., "Site-specific photochemistry along a protonated peptide scaffold," *J. Am. Chem. Soc.* 146, 13282–13295 (2024).